EMATHRIX EDUCATION · DATA SCIENCE & DATA ANALYTICS · WEEK 2

Automation & AI: Pipelines and Machine Learning basics

Read this first. It explains the concepts behind Week 2 — automation and AI — before you build anything. Basahin muna ito.

1 · Reading (you are here) → 2 · Activity 1 → 3 · Activity 2

1. What is automation? What is a pipeline?

Automation means a computer does a repetitive task for you, on a schedule, without you clicking anything. A **data pipeline** is a chain of automated steps that moves data from a *source* to a *destination*, transforming it along the way — for example: **fetch stock prices** → **score them** → **append to a Google Sheet** → **send an alert**.

*Tagalog: ang **pipeline** ay parang linya ng produksyon para sa data — kunin, ayusin, itago, mag-abiso — awtomatiko, walang manu-manong pindot.*

> **Key idea:** once a pipeline is built, data keeps arriving on its own. Your notebook then just *analyzes whatever the pipeline has collected*. Tools like **n8n** connect the steps without heavy coding.

2. What is a model (machine learning)?

A **model** is a program that *learns patterns from examples* instead of being told exact rules. You show it many labeled examples ("this is signed", "this is unsigned"); it adjusts itself until it can guess the label for a **new** image it has never seen. That guess comes with a **confidence** — how sure the model is (e.g. 92%).

*Tagalog: ang **model** ay natututo mula sa mga halimbawa, hindi mula sa mga patakarang isinulat mo. Bawat hula ay may **confidence** — gaano ito kasigurado.*

3. What does Teachable Machine do?

Teachable Machine (by Google) lets you train an image model **with no code**: you create classes, add example pictures, click **Train**, then **export** the model as a `.zip`. In Week 2 you upload that zip into Colab, run it over a folder of images, and let it classify each one and report its confidence.

> **Why confidence matters:** low-confidence guesses (say under 80%) should be sent to a human to double-check. AI assists judgement — it does not replace it.

4. Honest limits

- A model is only as good as its examples — biased or few examples give bad guesses.
- High confidence is **not** the same as being correct. Always sanity-check.
- Keep secrets safe: never paste real API keys into a shared notebook.

Proceed to the activities →